**LAB EXPERIMENT RECORD**

**Department: AIML**

**Course Code / Course Name: DSA0601 / Data Handling and Visualization**

**Experiment: Data Visualization using R and Tableau**

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**13-10-2026---Set 16: Product Inventory Management**

**Aim**

To analyze product inventory data using a bar chart, stacked bar chart and table, and to build an interactive Tableau dashboard for exploring the inventory data.

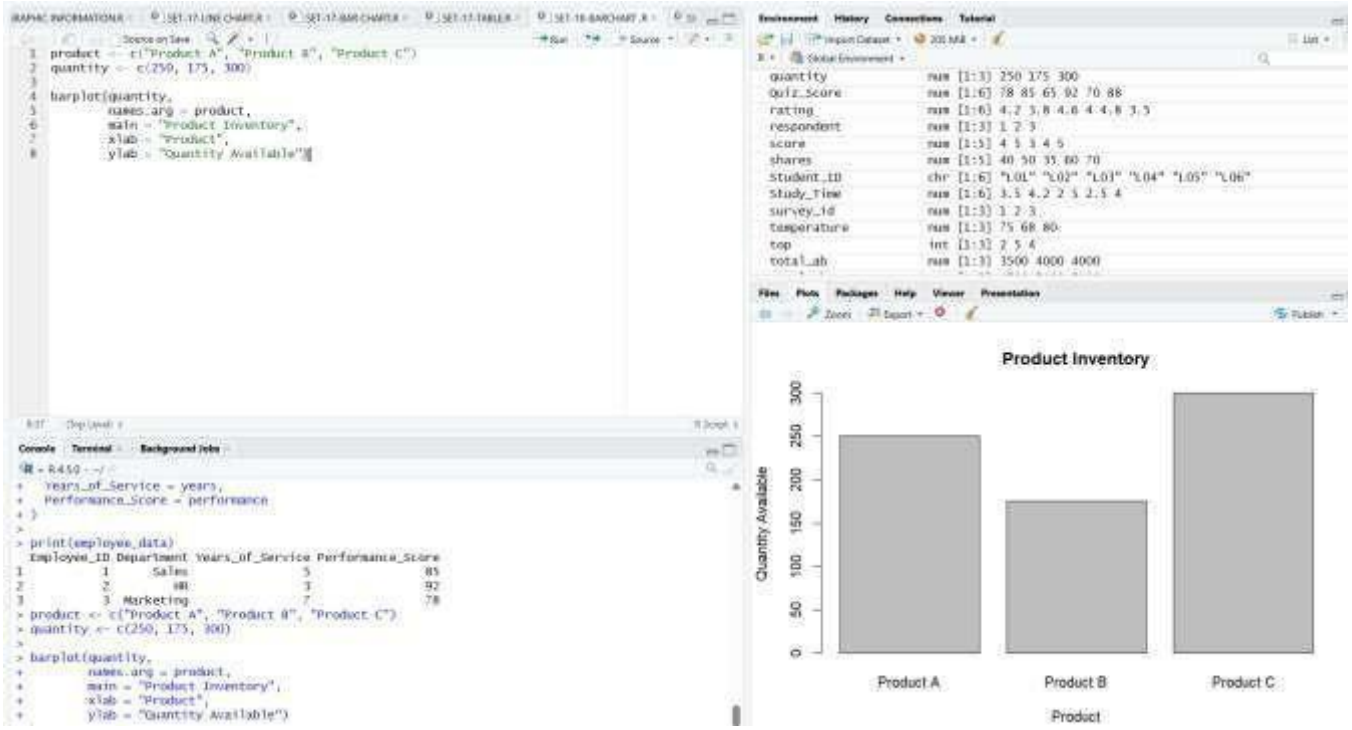
**Dataset: Product Inventory**

| Product ID | Product Name | Quantity Available | Price (in $) |
|---|---|---|---|
| 1 | Product A | 250 | 20 |
| 2 | Product B | 175 | 15 |
| 3 | Product C | 300 | 18 |

**Question 1: Bar Chart of Product Quantity**

**Question: Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.**

**Aim:** To visualize the quantity available for each product using a bar chart with labelled axes and a title.

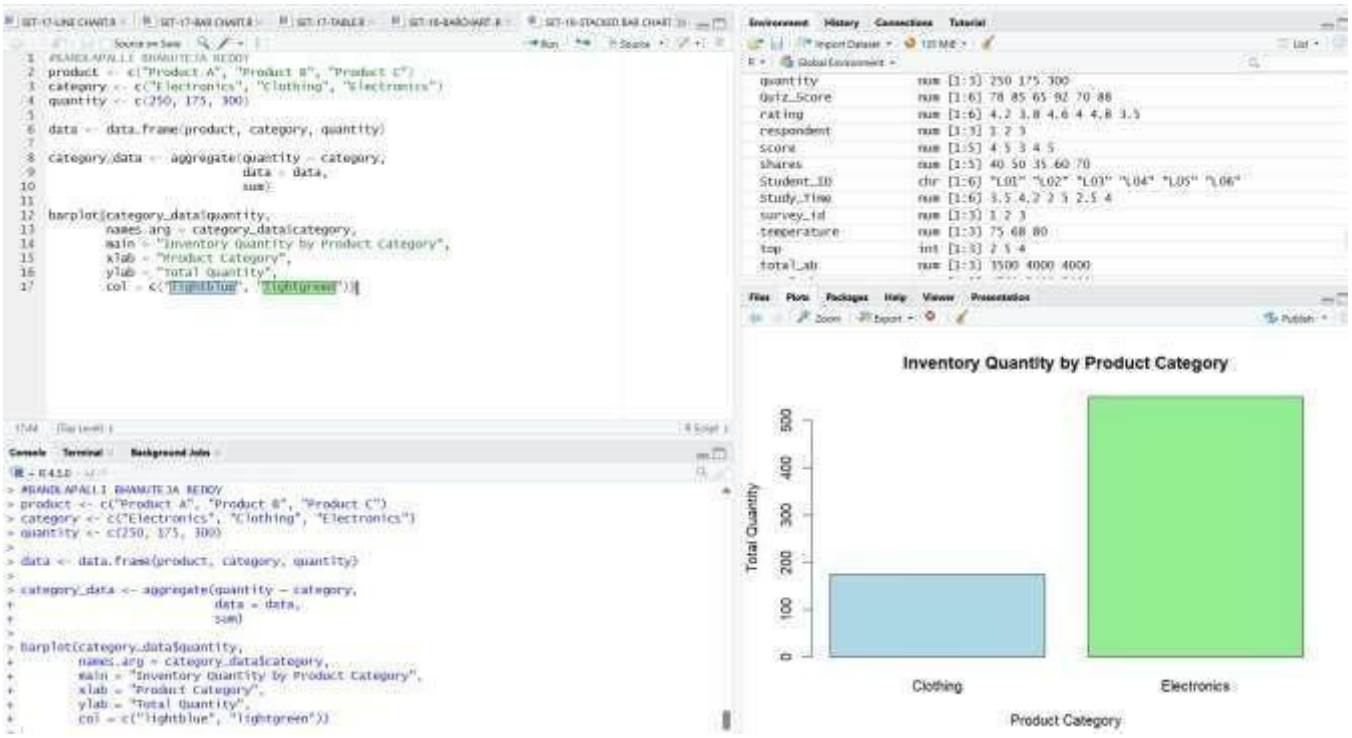



**Result:** Product C has the highest stock (300) and Product B the lowest (175), so Product B needs restocking first.

**Question 2: Stacked Bar Chart by Product Category**

**Question: Generate a stacked bar chart to show the quantity of each product within different product categories.**

**Aim:** To show how the quantity of each product is divided across product categories using a stacked bar chart.

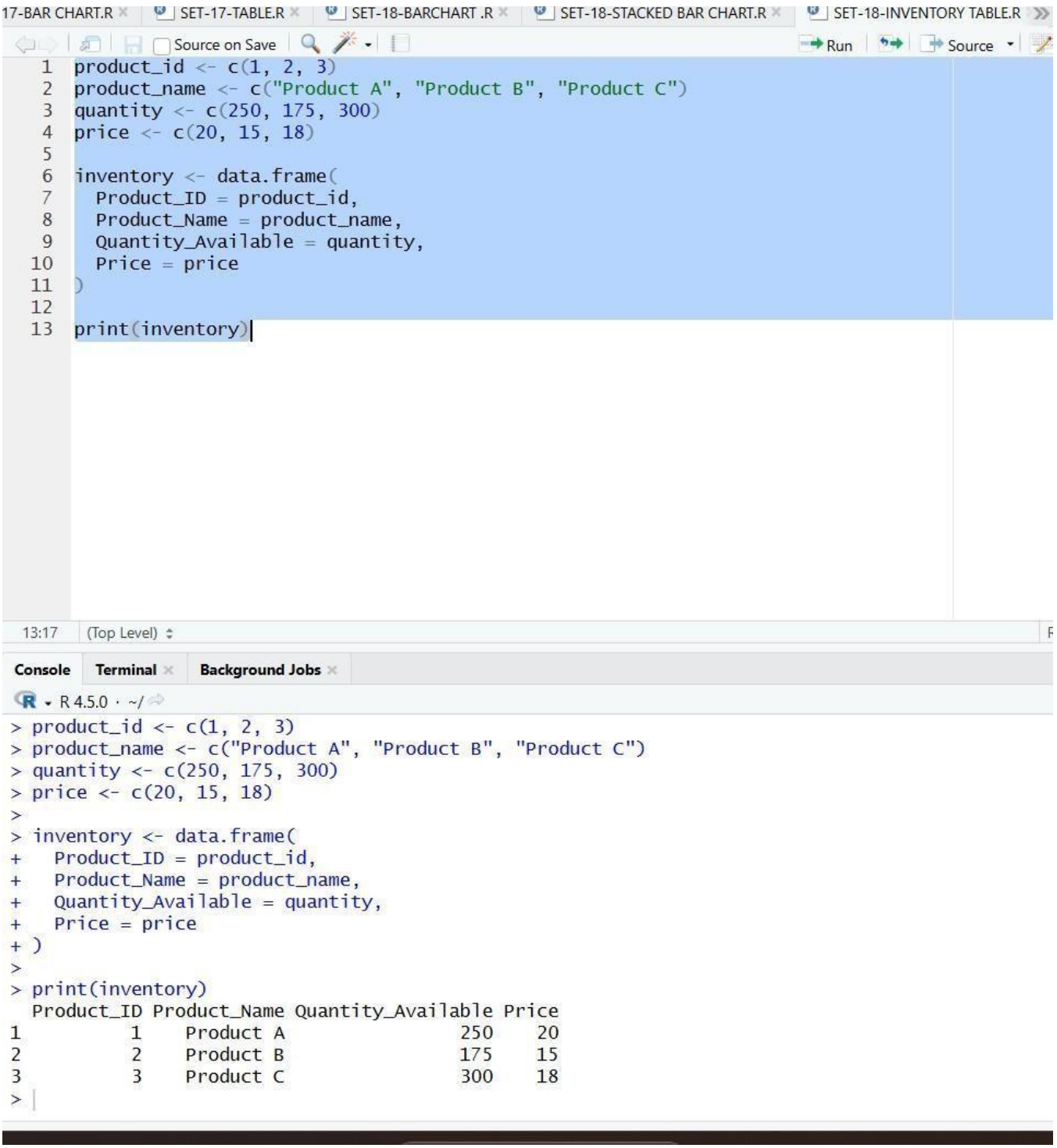



**Result:** Each bar is split by category, showing each category's share of a product's stock; the dataset gives no category column, so categories must be supplied.

**Question 3: Table of Inventory Data**

**Question: Build a table to show the inventory data for each product. Label the table elements.**

**Aim:** To present the quantity and price of each product in a clearly labelled table.




**Result:** The table lists all three products; the stock values are 5000 for Product A, 2625 for Product B and 5400 for Product C, so Product C holds the most value.

**Question 4: Interactive Tableau Dashboard**

**Question: Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.**

**Aim:** To build an interactive Tableau dashboard that combines the bar chart, stacked bar chart and table for exploring inventory data.




**Result:** The dashboard places all three views together, and a product filter updates them so users can explore the inventory interactively.

**Overall Result**

**Observation:** Total stock is 725 units with a total value of 13,025, and Product C is the largest item in both quantity (300) and value (5400).